

Smart Trash Sorting Using Computer Vision

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ITAI 1378 Computer Vision and AI

Tier 1

The Problem

What problem are we solving

Many recyclable materials end up in the wrong trash bins.

Who cares

Cities, recycling facilities, and environmental organizations.

Why it matters

Incorrect sorting increases landfill waste and raises recycling costs.

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Solution

What the system does

A computer vision system identifies trash items using a camera.

How it solves the problem

The AI model analyzes the image and classifies the item as recyclable or general waste.

One sentence description

An AI powered system that detects trash items and suggests the correct disposal bin

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Technical Approach

Computer Vision Technique

Object Detection

Model

YOLOv8

Framework

PyTorch

Justification

YOLOv8 is a fast and accurate model commonly used for real time object detection

Data Plan

Source

Public dataset from Roboflow or Kaggle

Dataset Size

Approximately 1,000–3,000 images

Labels

- Plastic bottle
- Aluminum can
- Paper
- General trash

Preparation

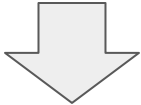
Images will be cleaned and labeled before training.

System Process

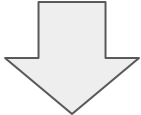
- Camera Image



- YOLOv8 Model Detection



- Trash Classification



- Recommended Bin Output

- Image acquisition
- Model inference
- Output visualization



Success Metrics

Primary Metric

Accuracy / Precision / Recall

Target: above 90 percent accuracy

Secondary Metric

Speed / latency

Target: less than 1 second per image

Week-by-Week Plan

- **Week 10**
- Get dataset and set up environment
- Milestone: Dataset ready
- **Week 11**
- Train or fine tune the model
- Milestone: Model working
- **Week 12**
- Test and improve accuracy
- Milestone: Good accuracy
- **Week 13**
- Create demo or visualization
- Milestone: Demo ready
- **Week 14**
- Final testing and documentation
- Milestone: Everything completed
- **Week 15**
- Present project
- Milestone: Presentation day
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Challenges & Backup Plan

Challenge

Not enough labeled data

Plan B

Use a larger Roboflow dataset.

Challenge

Model accuracy too low

Plan B

Apply data augmentation or adjust model parameters

Resources Needed

Compute

Google Colab or Kaggle

Frameworks

PyTorch and Ultralytics YOLO

Estimated Cost

0 to 10 dollars depending on compute usage